

Suspension Arm (Control Arm) - Structural Analysis Report

Note: This report is based on illustrative analysis images and is suitable for academic presentation purposes only.

1. Introduction

The suspension arm (control arm) is a critical component in an automotive suspension system. It connects the wheel assembly to the chassis and transfers loads generated during braking, cornering, and road impacts.

Model Description

The analyzed component consists of a central hub region connected to two mounting points through reinforced arms. Weight reduction pockets and fillets have been incorporated into the design.

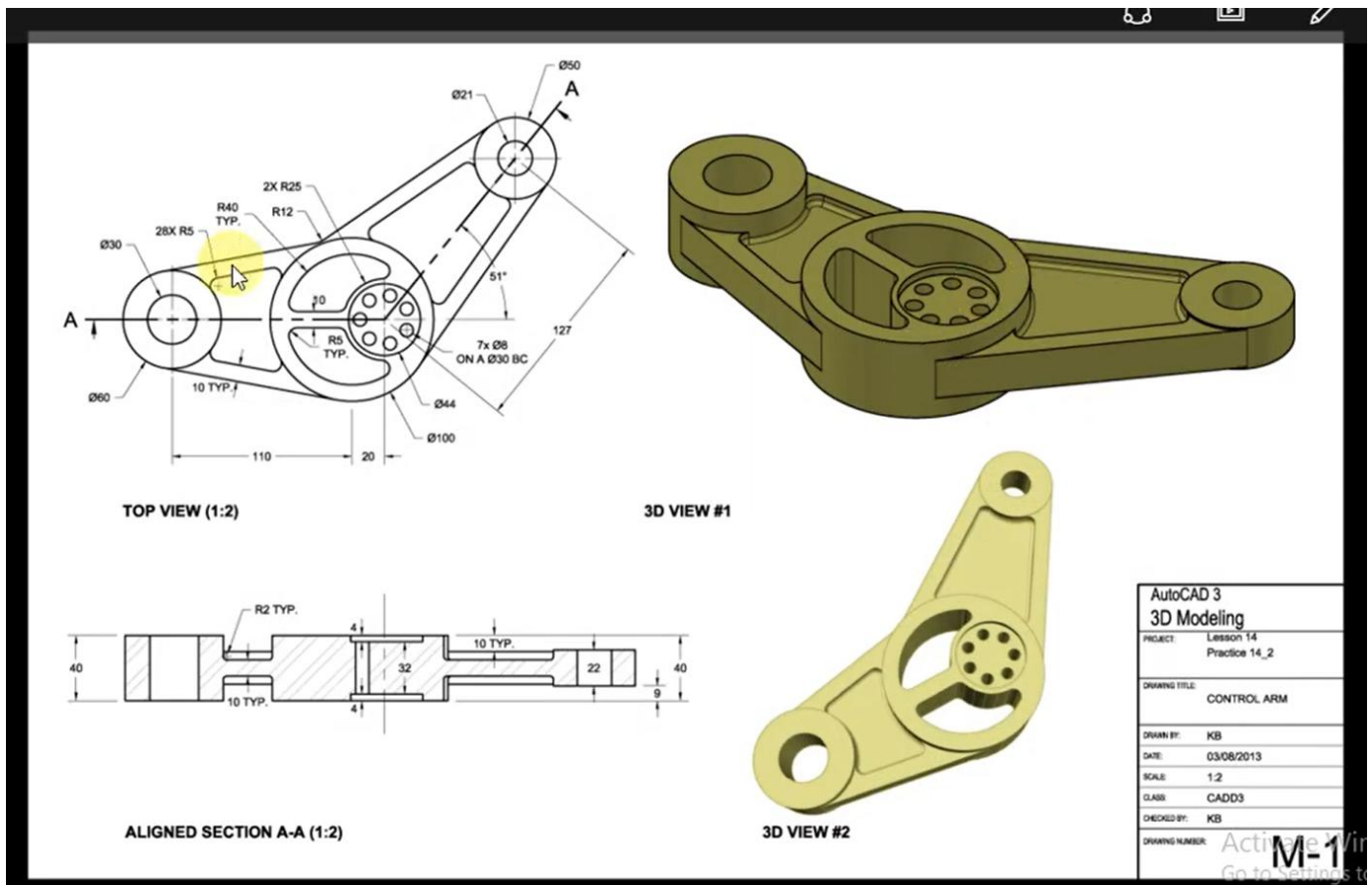


Figure 1.1 2D drawing of the suspension arm

2. Boundary Conditions and Loading

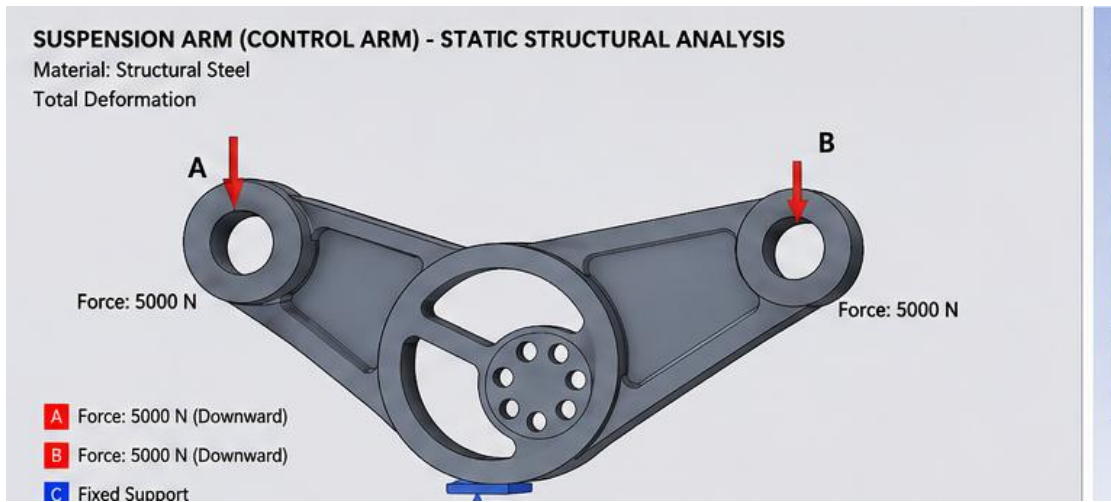


Figure 2.1 Static structural analysis

Two downward loads of 5000 N were applied at the outer mounting locations. The central mounting region was considered fixed to represent chassis attachment.

Assumptions

Linear static structural behavior, isotropic material properties, and small deformation assumptions were considered.

3. Results and Discussion

Total Deformation

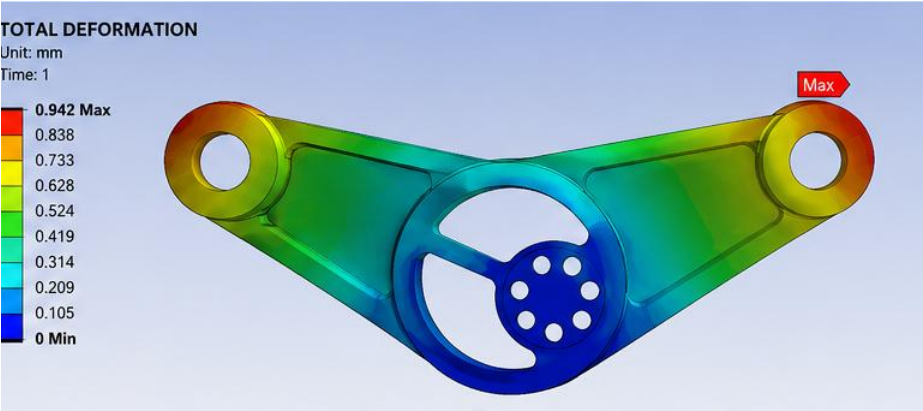


Figure 3.1 Total deformation

Von Mises Stress

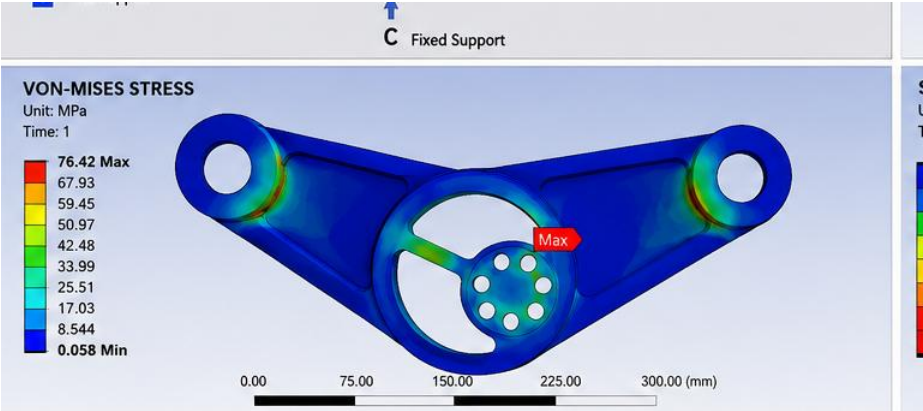


Figure 3.2 VON-Mises stress

Safety Factor

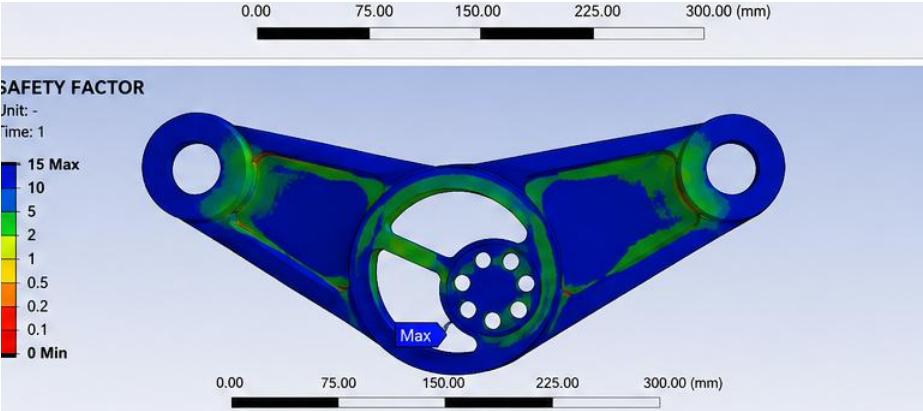


Figure 3.3 Safety factor

The highest deformation occurs near the loaded ends, while stress concentrations appear around geometric transitions and mounting regions.

4. Additional Results and Conclusion

Directional Deformation

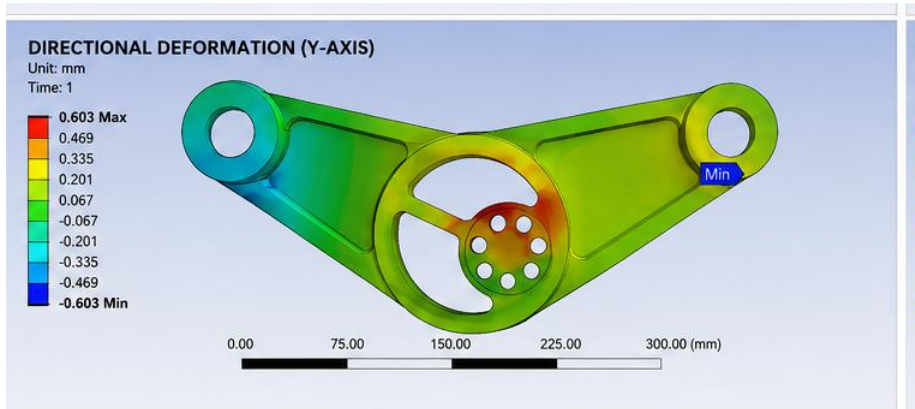


Figure 4.1 Directional deformation

Equivalent Elastic Strain

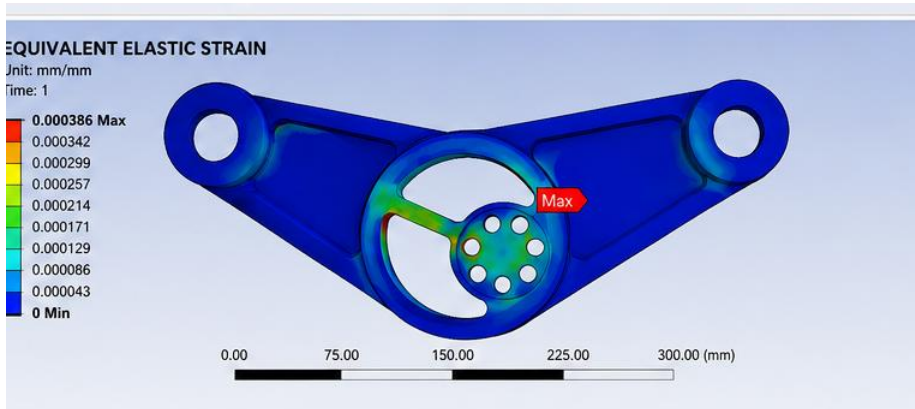


Figure 4.2 Equivalent Elastic strain

Conclusion

The suspension arm demonstrates acceptable structural behavior under the applied loading conditions shown in the illustrative analysis. Maximum stress and strain are concentrated near mounting and transition regions. Future work may include fatigue analysis, dynamic loading, and optimization for weight reduction.